ROOT ZONE ENVIRONMENT MONITORING FOR SUSTAINABLE ANTHURIUM CULTIVATION USING INTELLIGENT ANALYSIS

Project Id: 25-26J-357

Project Proposal Report

GAMAGE P.U.
IT22131188

BSc (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
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DECLARATION

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We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any university or institute or higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

....29/8/25.....

Signature of Supervisor Date

(Mrs. Anjalie Gamage)

ABSTRACT

Anthurium, a high-value ornamental plant, demands precise soil conditions to achieve optimum growth and yield. However, traditional soil management practices are often limited to periodic manual testing, which provides generalized insights and fails to capture the dynamic nature of soil health. Farmers face challenges such as moisture imbalances, nutrient deficiencies, and temperature-related stress in the root zone, leading to reduced quality and productivity. This research proposes the Root Zone Environment Monitoring Module,designed to continuously monitor and analyze critical soil parameters such as moisture, pH, nutrient content, and temperature. By integrating soil sensors with GPS-based geolocation mapping, the module provides real- time, location-specific soil health data across the cultivation area. The collected data is processed through an intelligent analysis engine that compares readings against ideal Anthurium soil requirements, enabling early detection of imbalances and generating actionable recommendations. These include adjustments to irrigation,fertilizer application, or planting zone selection, supporting proactive rather than reactive farming. The system also visualizes spatial variations in soil quality through geospatial maps, assisting cultivators in making informed, data-driven decisions. Unlike conventional methods, this approach combines continuous monitoring with predictive insights, ensuring sustainability and resource efficiency. In essence, the Root Zone Environment Monitoring Module bridges the gap between traditional soil analysis and intelligent decision support by delivering precise, real-time recommendations for Anthurium cultivation. It contributes to sustainable agriculture by improving crop quality, conserving resources, and empowering farmers with actionable insights.

Contents

DECLARATION i

ABSTRACT ii

LIST OF FIGURES iv

LIST OF TABLES v

LIST OF APPENDICES vi

LIST OF ABBREVIATIONS vii

1. INTRODUCTION 1

1.1 Background & Literature survey 1

2. OBJECTIVES 6

2.1 Main Objectives 6

2.2 Specific Objectives 6

3. METHODOLOGY 8

4. PROJECT REQUIREMENTS 11

4.1 Functional requirements 11

4.2 Non-functional requirements 11

5. GANTT CHART 14

6. WORK BREAKDOWN CHART 15

7. BUDGET AND BUDGET JUSTIFICATION 16

8. REFERENCES 17

9. APPENDICES 18

LIST OF FIGURES

Figure 1 -- System Architecture 7

Figure 2 -- Gannt Chart 11

Figure 3 -- Work breakdown Chart 12

LIST OF TABLES

Table 1 --- Budget Table 14

LIST OF APPENDICES

Appendix 1 -- Plagiarism Report 16

LIST OF ABBREVIATIONS

Abbreviation	Meaning
IoT	Internet of Things
GPS	Global Positioning System
ML	Machine Learning
UI	User Interface
DB	Database
SaaS	Software as a Service
API	Application Programming Interface
WSN	Wireless Sensor Network
GIS	Geographic Information System
RS	Remote Sensing
CSV	Comma-Separated Values (data format)

1. INTRODUCTION

1.1 Background & Literature survey

Sustainable cultivation of high-value ornamental crops, such as Anthurium, demands careful monitoring of environmental and soil conditions to ensure optimal growth, flowering, and yield [1]. Traditional soil management practices, including manual soil sampling and periodic laboratory testing, provide limited insights into spatial variations across cultivation fields and fail to support timely, data-driven decisions [2]. As a result, farmers often adopt reactive strategies, which can lead to nutrient deficiencies, moisture imbalances, and reduced plant health [3].

Critical soil characteristics like moisture, pH, temperature, and nutrient content can now be continuously monitored thanks to developments in data-driven agriculture [4][5]. Farmers can identify possible stress situations before they have a detrimental effect on crop performance by gathering high-frequency, location- specific data. These data, when paired with geographic mapping, can be represented as maps of soil health, allowing for more focused treatments like better fertilization, irrigation, and soil amendments [6].

Despite these advancements, the majority of current solutions are either crop-specific tools or generic farm management systems. Few use AI-based analysis to forecast nutrient deficits or soil stress and offer prescriptive suggestions based on crop-specific needs [7]. Moreover, present methods hardly ever integrate real-time and historical data to provide proactive recommendations for enhancing flowering and yield results.

A GeoSoil Monitoring Module that continuously assesses soil conditions, compares them to ideal ranges for a given crop, detects possible stressors, and produces practical suggestions for soil management, fertilization, and irrigation is the subject of this study. In the end, the system improves plant health, flowering, and yield while conserving resources by utilizing data analysis, predictive modeling, and geographic visualization to enable proactive, accurate, and sustainable crop management [1][4][6].

1.2 Research Gap

High-value decorative crops like Anthurium still have little AI integration in real-time soil monitoring and decision assistance, despite the advancement of precision agriculture technologies. Current systems lack
localized, sensor-driven soil analysis and predictive recommendations, but they primarily offer generic weather updates or simple dashboards. The standard practice is still manual soil testing and sporadic lab results, which are frequently delayed, expensive, and inadequate for proactive farming techniques..

Feature	Current Solutions	Proposed Solutions
Soil Parameter Monitoring	Manual or periodic lab testing	Real-time monitoring of moisture, pH, temperature, and nutrients using sensors
Geospatial Analysis	Limited or no location-specific insights	GPS-integrated mapping showing soil health variations across the field
Predictive Recommendations	Farmer intuition or static schedules	AI-driven analysis generating actionable interventions and alerts
Resource Optimization	Generalized irrigation/fertilization	Data-driven, location-specific recommendations to optimize water, fertilizer, and energy usage

Table 2: Research gap analysis

High-value decorative crops like Anthurium continue to encounter difficulties with data-driven cultivation and real-time soil monitoring, despite developments in precision agriculture and IoT technology [1][2]. The majority of current methods depend on labor-intensive manual soil testing or sporadic laboratory analysis, neither of which can offer localized insights for proactive decision-making [3]. These restrictions frequently lead to ineffective resource use, nutritional shortages, and less than ideal flowering.

Farmers are unable to comprehend field-specific differences in soil quality because existing digital solutions rarely combine geospatial mapping with sensor networks [4]. The majority of technologies only offer aggregate data, which restricts the use of precision treatments such soil amendments, targeted irrigation, and fertilization [5].

These gaps are filled by the proposed GeoSoil Monitoring Module, which combines GPS-integrated mapping, AI-powered analysis, and IoT-based soil sensors. In order to improve crop health and productivity, it offers location-specific insights, identifies departures from ideal conditions, and produces practical suggestions for soil management, fertilization, and irrigation.

1.3 Research Problem

High-value decorative plants like anthuriums need very particular soil types, moisture content, and environmental conditions in order to flourish. However, manual observations, farmer intuition, and generic or rare soil testing are frequently used in traditional agriculture methods. Localized soil deficiencies, moisture imbalances, temperature swings, and nutrient shortages that can have an immediate impact on plant health, flowering, and production cannot be detected with these techniques. Because of this, growers frequently experience erratic results, decreased output, and monetary losses..

The cultivation of Anthurium is only partially supported by current digital farming solutions. Real-time soil monitoring and geographic data mapping are not integrated by the majority of platforms, which instead concentrate on providing basic farm management dashboards or generic weather updates. This leads to a crucial gap: farmers are unable to monitor changes in soil health over time or pinpoint particular areas in their fields that might need quick attention. Allocating resources, such as irrigation, fertilization, or soil amendments, remains inefficient in the absence of ongoing, location-specific insights, frequently resulting in excessive or insufficient usage of water, nutrients, and fertilizers..

The lack of sophisticated decision-support systems that convert sensor data into useful advice is another significant obstacle. Temperature, pH, nutrient content, and soil moisture can all be measured by sensors, but raw data is not enough. A system that can analyze data, spot trends, forecast possible hazards, and suggest focused treatments is essential for farmers. Growers are left to maintain ideal cultivation conditions reactively rather than proactively via traditional methods, which do not offer predictive insights based on past and current soil conditions.

Moreover, it is uncommon for existing systems to combine several data sources into a coherent system. Though it is yet mostly unexplored for Anthurium farming, the combination of soil sensors, GPS mapping, weather data, and historical trends is crucial for precision agriculture. Farmers' capacity to apply data-driven strategies is diminished by the absence of such integrated systems, endangering efficiency and sustainability.

In order to overcome these constraints, this project suggests a GeoSoil Monitoring Module that

uses sophisticated soil sensors, GPS-based geospatial mapping, and sophisticated analysis algorithms to deliver location-specific, real-time soil information. The module will track past trends for predictive decision-making, identify deviations in soil characteristics, and produce practical recommendations. By doing this, the system hopes to provide growers with proactive, data-driven management that will improve plant health, increase productivity, and promote sustainable Anthurium farming methods.

2. OBJECTIVES

2.1 Main Objectives

The primary goal of this research is to develop a data-driven decision support system for sustainable Anthurium cultivation that continuously monitors root-zone soil conditions and provides actionable recommendations to optimize plant health, flowering, and yield. The system focuses on collecting key soil parameters such as moisture, pH, temperature, and nutrient levels across different cultivation zones. By analyzing these readings against crop-specific optimal ranges, the system can detect nutrient deficiencies, moisture imbalances, or temperature-related stress, enabling farmers to take timely and precise corrective actions [1][4][6].

To improve decision-making, the system incorporates historical data trends and predictive analysis in addition to real-time monitoring. The software facilitates proactive and sustainable farming practices by displaying geolocation-based soil health maps and producing tailored irrigation, fertilization, and soil management suggestions. The ultimate goal is to develop a clever, intuitive tool that gives farmers the ability to enhance crop quality, save resources, and boost output by providing them with data-driven insights and practical advice.

2.2 Specific Objectives

- Design a soil data acquisition framework to collect real-time parameters such as soil moisture, pH, temperature, and nutrient levels from Anthurium cultivation zones.

- Develop a central database and analytics pipeline to process and store soil health data, ensuring accuracy, scalability, and secure access for decision-making.

- Implement soil condition classification models to identify imbalances or

stress factors affecting Anthurium growth and recommend corrective measures.

- Create geolocation-based visualization dashboards that allow farmers to view soil condition variations across cultivation zones for better resource allocation.

- Integrate a recommendation engine to provide actionable insights on irrigation scheduling, fertilization, and soil management tailored to Anthurium's growth stages.

- Establish a feedback and performance evaluation loop to validate system recommendations with actual crop outcomes, continuously improving accuracy and reliability.

3. METHODOLOGY

3.1 System Architecture

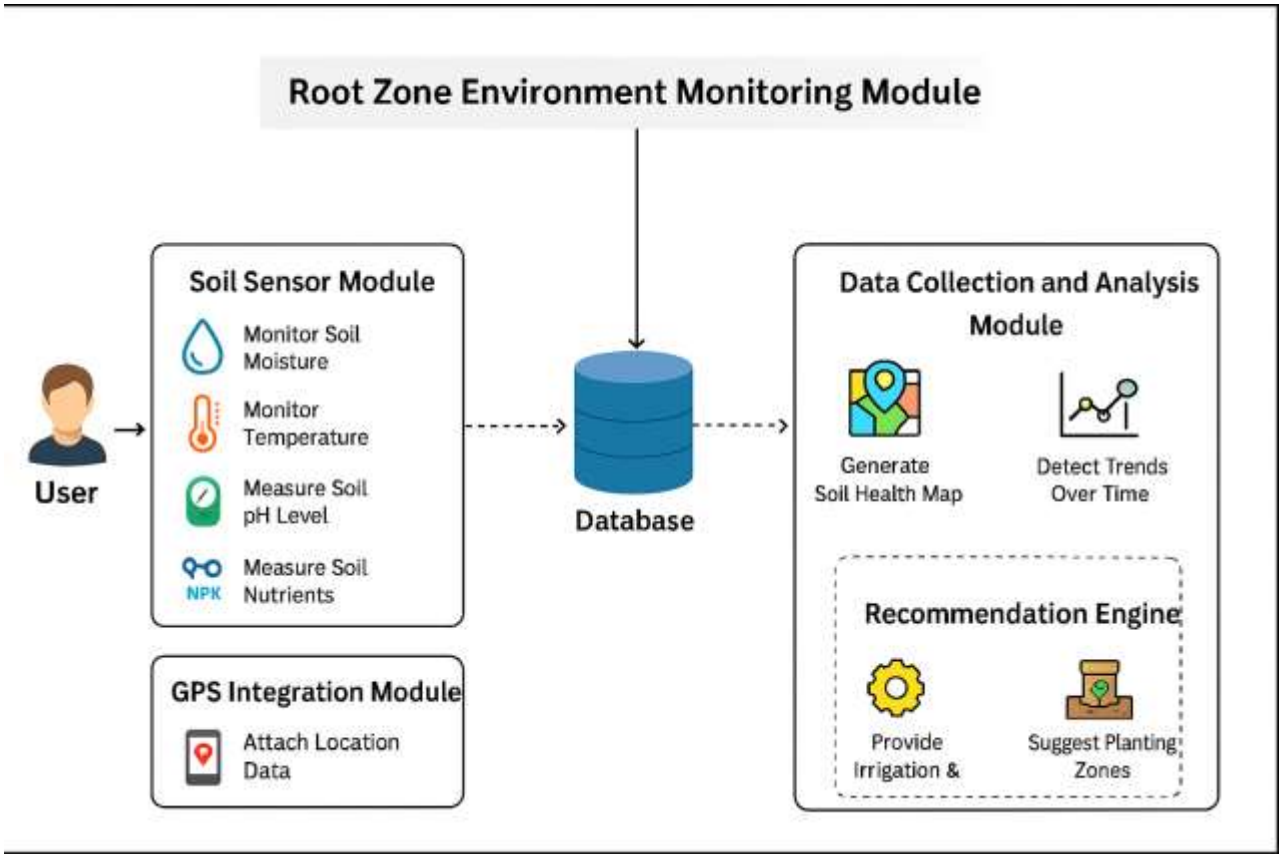

Figure 1 -- System Architecture

3.1.1 Software Solution

Tools and Technologies

- Flutter
- Python
- TensorFlow
- Flask
- Google Colab
- Firebase Firestore
- Weather APIs
- Google Maps SDK

3.1.2 Commercialization

- Both in Sri Lanka and internationally, the Root Zone Environment Monitoring Module has a great chance of becoming a commercial product in the precision agriculture and smart farming sectors.
- Target Audience:
    - commercial flower farmers and anthurium growers.
    - agronomists and agricultural research organizations.
    - suppliers of IoT solutions and smart farming technologies.
- Market Opportunity:
    - The global Ayurvedic market was valued at USD 9.5 billion in 2023 and is projected to grow at 15% CAGR until 2030.
    - The global AI in healthcare market is expected to surpass USD 200 billion by 2030, indicating strong integration opportunities.
- Revenue Models:
    - Sensor kits with cloud monitoring subscriptions are sold as part of a hardware and software package
    - Providing the platform to commercial farms and nurseries is known as business-to-business licensing.
    - SaaS Model: subscriptions for cloud-based soil monitoring and analytics for various crop kinds.
    - Consulting & Data Insights: offering farms studies on soil optimization and predictive analytics.

- Competitive Advantage:
    - replaces recurring testing with real-time, location-specific soil insights.
    - combines GPS mapping, sensors, and cognitive analysis to provide recommendations that may be put into practice.
    - enhances output and resource efficiency by assisting farmers in switching from reactive to proactive farming..
- Business Feasibility:
    - Strengths: Innovative, useful, adaptable to other crops, and consistent with advancements in precision agriculture.
    - Weaknesses: Requires farmer training and an initial sensor expenditure.
    - Opportunities: value crops, government incentives for technological adoption, and the growing use of smart farming.
    - Threats: conventional farming practices that are reluctant to incorporate new technologies and competition from generic soil monitoring systems.

4. PROJECT REQUIREMENTS

4.1 Functional requirements

- Image capture and analysis: Allow users to upload or capture images through the mobile app for processing using CNN-based models.
- Location-based insights: Use Google Maps API with geolocation tagging to provide contextual results and personalized recommendations.
- Rule-based decision support: Apply custom rule-based logic along with Decision Trees to refine predictions and recommendations.
- Real-time notifications: Deliver push notifications for reminders, alerts, and location-specific updates.
- User profile management: Enable secure storage of user profiles, preferences, and history in Firebase for personalized experiences.
- Report generation: Automatically generate structured reports summarizing analysis results and recommendations.

4.2 Non-functional requirements

- Cross-platform compatibility: Ensure smooth operation on both Android and iOS devices through Flutter.
- Scalability: Handle growing numbers of users and datasets efficiently without service disruption.
- Performance optimization: Provide low-latency responses for image analysis and location-based processing.
- Data security: Encrypt sensitive data and ensure secure authentication via Firebase.
- Maintainability: Use modular architecture to simplify future feature integration (e.g., advanced deep learning models).

4.3 User requirements

- Consent management: Obtain user permission before collecting images, location data, or personal information.
- Personalized recommendations: Provide lifestyle or contextual suggestions based on image analysis, geolocation, and rule-based logic.
- Accessibility: Ensure a simple and intuitive mobile interface suitable for users with varying technical skills.
- Progress tracking: Store user activity and generate history-based insights for better decision- making.
- Exportable reports: Allow users to download or share analytical reports in standard formats.

4.4 Expected Test Cases

Test Case	**Input**	**Expected Output**	**Metric**
Soil Sensor Data Collection	Real-time readings from moisture, pH, temperature, nutrient sensors	Accurate and complete soil data captured	Data accuracy ≥90%
GPS Mapping	Sensor location coordinates	Correct geolocation mapping on soil health map	Detection accuracy ≥85%
Data Analysis	Collected soil data	Identifies nutrient deficiencies, moisture imbalance, or temperature stress	Relevance rating ≥4/5
Recommendation Generation	Analyzed soil data	Actionable recommendations for irrigation/fertilizer/planting	Trend detection success ≥80%
Historical Data Comparison	Past and current soil data	Detects long-term trends and potential future soil issues	Latency benchmark

Sensor Failure Handling	Sensor malfunction or missing data	Alerts user and marks missing data	Error handling success
Multi-Field Monitoring	Multiple fields or zones monitored simultaneously	Correct data collection and recommendations for each zone	Data completeness ≥90%
System Response Time	Large dataset from all sensors	Recommendations generated within acceptable time	Response time <3s

Table 4: Expected Test Cases

5. GANTT CHART

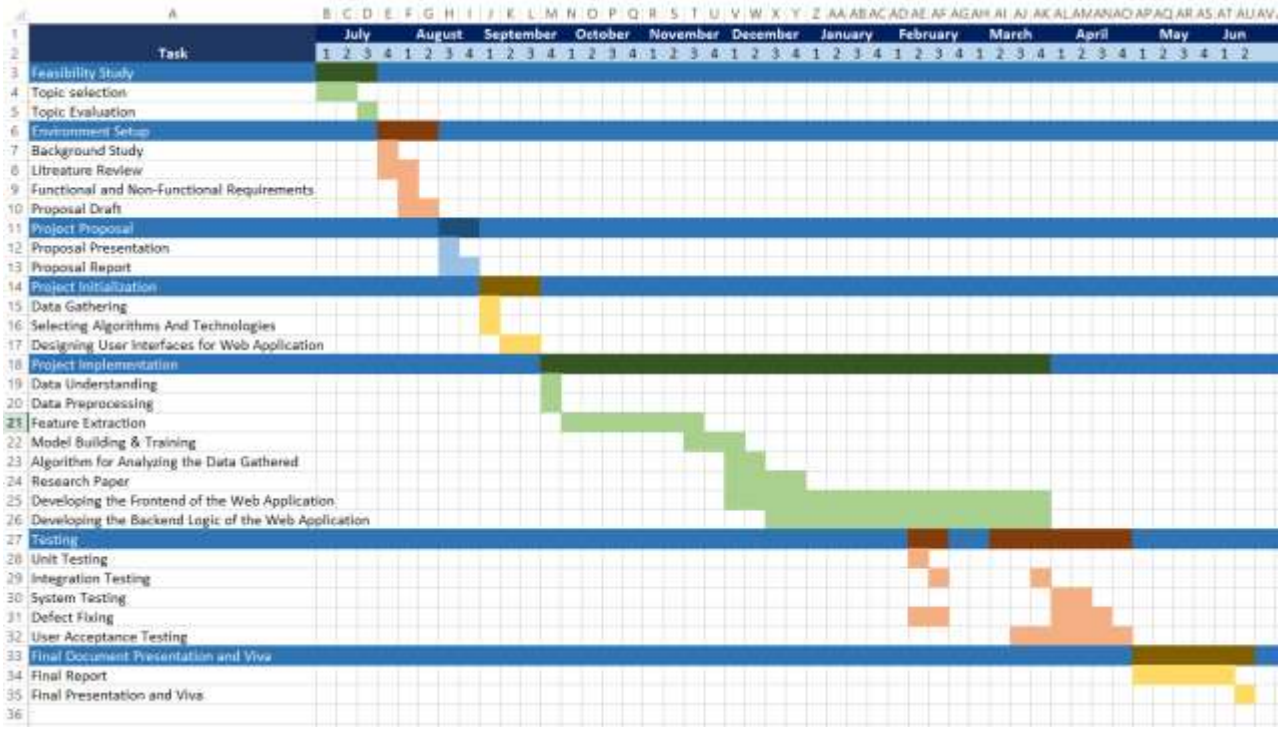

Figure 2 -- Gannt Chart

6. WORK BREAKDOWN CHART

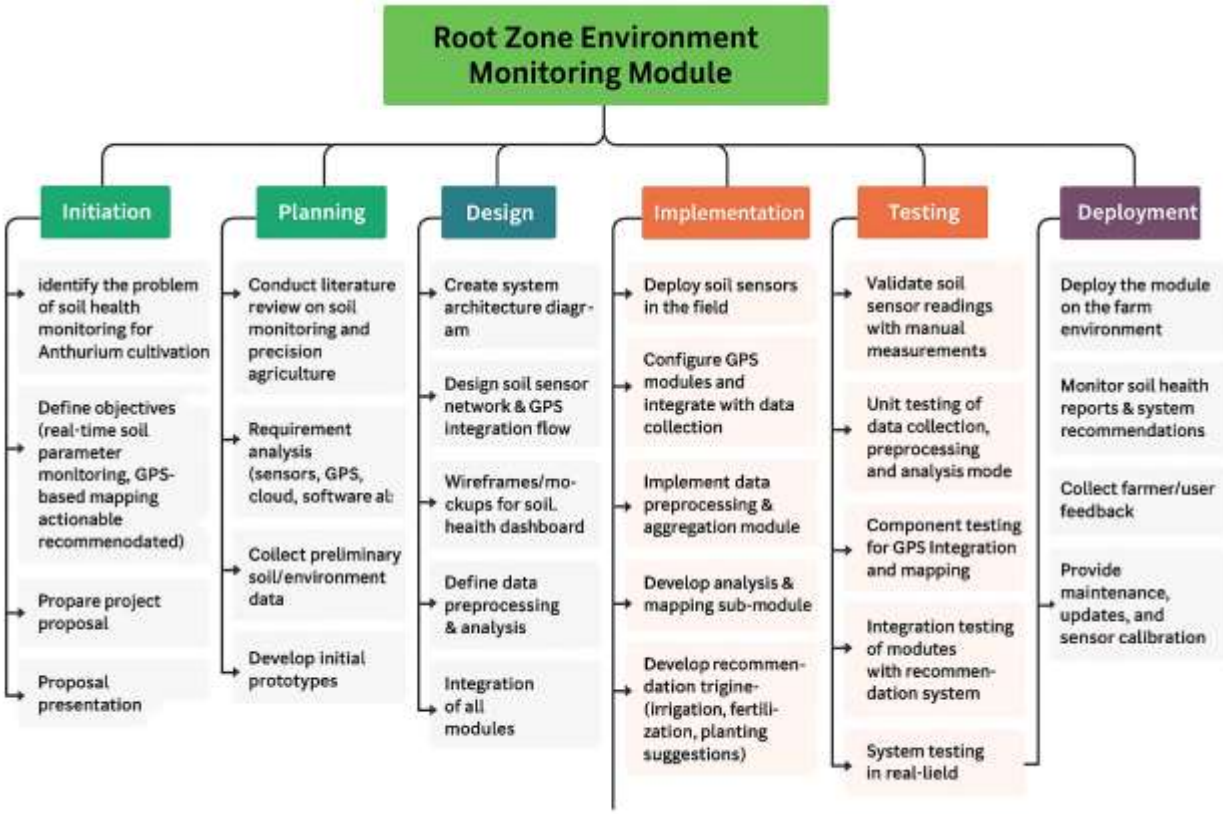

Figure 3 -- Work breakdown Chart

7. BUDGET AND BUDGET JUSTIFICATION

For this research component, we do not aim to commercialize our individual components and the whole research will be commercialized as an application. Following is the budget justification for the whole research system.

Description	Cost
Cost for Deployment	10000.00
Internet	4000.00
Travelling cost for data collection	4000.00
Google Map API	2000.00
Paper Publishing	20000.00
Total	40000.00

Table 1 --- Budget Table

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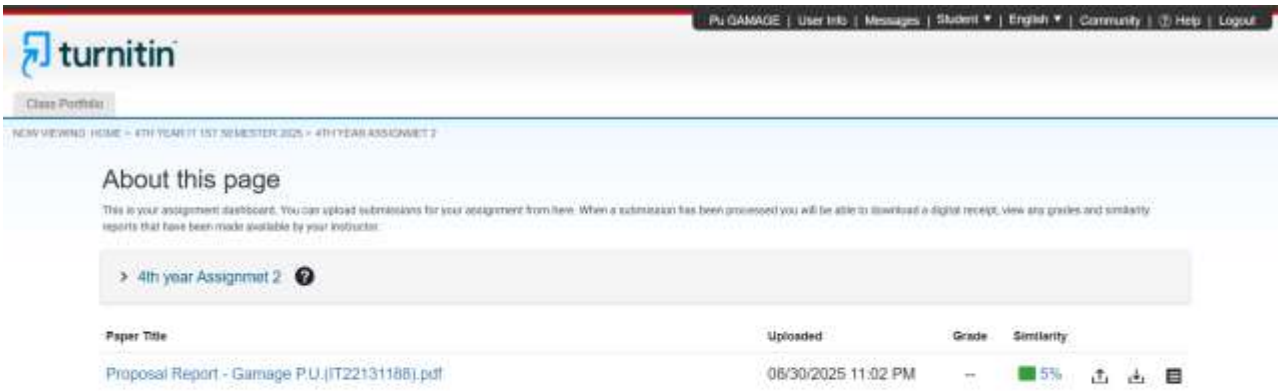
9. APPENDICES